Domain Research: Dataset for Multi-Turn Coding Agents

This research focuses on constructing a high-quality dataset for training language models to perform structured tool usage in multi-turn coding environments. The objective is to move beyond prompt-based tool invocation and instead embed tool interaction behaviour directly into model representations, enabling more autonomous and reliable coding agents.

Initial experimentation utilized existing tool-use datasets such as ToolBench and Hermes. These datasets demonstrated that models can learn to follow system-level JSON formatting constraints with reasonable accuracy. However, a consistent limitation was observed: models tend to assume or hallucinate tool names rather than strictly adhering to the provided schema. This behavior is attributed to ambiguity and inconsistency in tool naming across these datasets, resulting in weak alignment between tool identifiers and their intended functions.

To address the limitations of existing datasets, synthetic data generation was explored using a 32B Qwen2.5 Instruct model. While the model was capable of producing structured outputs, it exhibited significant weaknesses in generating coherent multi-turn trajectories. Specifically, it struggled with maintaining consistency across steps, producing realistic iterative workflows, and preserving logical continuity between tool calls and their outcomes. As a result, the generated data lacked the depth and reliability required for training effective coding agents.

Following this, the focus shifted to the SWE-chat dataset, which consists of real-world interactions between developers and AI systems. This dataset inherently captures authentic multi-step workflows, including file inspection, code modification, command execution, and iterative debugging cycles. Unlike synthetic or templated datasets, SWE-chat provides naturally occurring patterns of failure, reasoning, and validation. However, it is also large, noisy, and contains substantial redundancy, requiring careful processing to extract meaningful training data.

The current research direction emphasizes transforming raw interaction logs into structured trajectories suitable for model training. This involves tool normalization, session reconstruction, trajectory compression, and goal inference. Key challenges include removing redundant steps, filtering low-signal outputs, maintaining meaningful iteration patterns, and ensuring consistency in tool representation.

This work contributes to the broader investigation of whether tool usage can be learned as an intrinsic capability of language models rather than being imposed through prompting. The long-term goal is to enable smaller models, such as those in the 3B parameter range, to perform complex coding tasks through learned multi-turn tool interactions.